

# DISORDERS OF LIVER IN CHILDREN

----- Active space -----

## Hereditary Hyperbilirubinemias

00:00:15

AKA Syndromic jaundice.

AR inheritance.

|           | ↑ unconjugated bilirubin (BR)                                                                                | ↑ conjugated bilirubin                                                                                 |
|-----------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Cause     | Unconjugated BR<br>$\downarrow$ UDP-GT $\rightarrow$ Defect $\rightarrow$ ↑ unconjugated BR<br>Conjugated BR | Defect in secretion/<br>excretion of conjugated BR<br>from liver into bile                             |
| Disorders | <ul style="list-style-type: none"> <li>• Crigler Najjar syndrome.</li> <li>• Gilbert syndrome.</li> </ul>    | <ul style="list-style-type: none"> <li>• Dubin Johnson syndrome.</li> <li>• Rotor syndrome.</li> </ul> |

### CRIGLER NAJJAR SYNDROME :

Suspect type I if there is high bilirubin without hemolysis beyond 1<sup>st</sup> week after birth. Khanilrao0392@gmail.com

|                                                                | Type I                                                                                                                                                   | Type II                                                                                                                 |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| UDP-GT                                                         | Absent                                                                                                                                                   | Partial deficiency                                                                                                      |
| Onset                                                          | Neonatal jaundice                                                                                                                                        |                                                                                                                         |
| Severity                                                       | most severe (Can be fatal)                                                                                                                               | Less severe                                                                                                             |
| Risk of kernicterus                                            | High                                                                                                                                                     | Low                                                                                                                     |
| Serum BR                                                       | > 20 mg/dl                                                                                                                                               | < 20 mg/dl                                                                                                              |
| management                                                     | Supportive :<br><ul style="list-style-type: none"> <li>• Phototherapy (PT).</li> <li>• Exchange transfusion (ET).</li> </ul> Curative : Liver transplant | Only supportive :<br><ul style="list-style-type: none"> <li>• Phototherapy.</li> <li>• Exchange transfusion.</li> </ul> |
| Upon administration of 5 mg/kg phenobarbitone (Enzyme inducer) | No response                                                                                                                                              | ↓ jaundice                                                                                                              |

### GILBERT SYNDROME :

m/c cause of hereditary hyperbilirubinemia.

Pathology : mild UDP-GT deficiency  $\rightarrow$  m/c benign type of hyperbilirubinemia.

Onset : Adolescence/puberty.

mild unconjugated BR (1-6 mg/dl).

104

## Gastrointestinal System

----- Active space -----

## Symptoms :

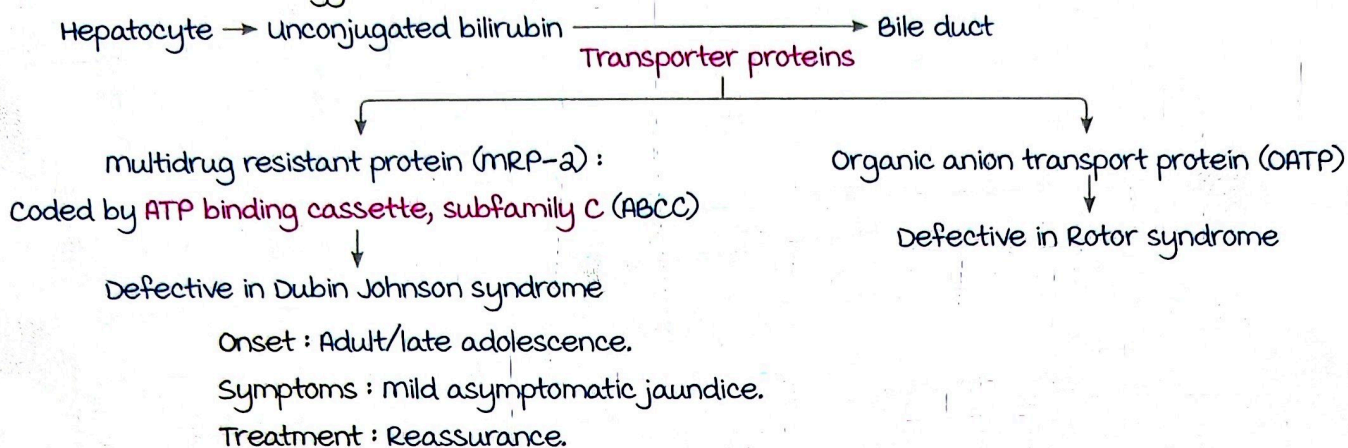
- Intermittent fluctuant jaundice.
- Any stressor (Illness/fasting) → Precipitates jaundice.

## Treatment :

- Reassurance.
- No intervention needed.

## DUBIN-JOHNSON AND ROTOR SYNDROME :

## Pathology :



## Neonatal Cholestasis

00:15:09

Impairment of bile flow in hepatobiliary system → ↑ conjugated BR.

## DEFINITION :

Depends on total serum bilirubin (TSB) values.

Biochemically :

Clinically :

- Persistent jaundice > 3 weeks (With exclusion of other diagnosis).
- Pale/clay (Especially the core) coloured stools.
- Dark yellow urine.

## ETIOLOGY :

## Intrahepatic :

1. Idiopathic neonatal hepatitis (2<sup>nd</sup> m/c cause).
2. Infections : TORCH, sepsis.
3. Metabolic causes :

648c85cfee3b03a74e182fab

a. Galactose-1-phosphate uridyl transferase deficiency :

- Oil drop cataract.
- Jaundice.

b. Tyrosinemia type I :

- Deficiency of fumaryl acetoacetate hydrolase.
- Boiled cabbage odour of urine.
- Hepatorenal syndrome : Proximal tubular damage → Fanconi like syndrome.

## c. Lysosomal storage disorders :

----- Active space -----

|                   | Gaucher disease                                                                                                                                                                           | Neiman pick disease                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Deficient enzyme  | Glucocerebrosidase                                                                                                                                                                        | Sphingomyelinase                                                                                                  |
| Clinical features | <ul style="list-style-type: none"> <li>Hepatosplenomegaly.</li> <li>Lytic bone lesions → Bone pain</li> <li>X-ray : <b>Erlen meyer flask deformity.</b></li> <li>Pancytopenia.</li> </ul> | <ul style="list-style-type: none"> <li>Hepatosplenomegaly.</li> <li><b>Cherry red spots</b> in macula.</li> </ul> |
| HPE               | Crumpled tissue paper appearance                                                                                                                                                          | Foamy cells                                                                                                       |

## 4. Syndromes :

**Alagille Syndrome :**

- Gene defects : JAGGED 1 & NOTCH 2.
- ↓ intrahepatic bile ducts
- Pulmonary artery stenosis } **Arteriohepatic dysplasia**
- **Triangular facies.**
- vertebral defects : Hemivertebra & butterfly vertebra.
- **Posterior embryotoxon** : Slit lamp finding.
  - white gray ridge at the cornea, near the limbus (Seen under slit lamp).

**Zellweger Syndrome :** Peroxisomal disorders.

- Gene defect : PEX → Codes for peroxins.
- Accumulation of very long chain fatty acids.
- Can be **misdiagnosed** as Down's syndrome :
  - Have similar facies.
  - Profound hypotonia.

**Extrahepatic causes :**

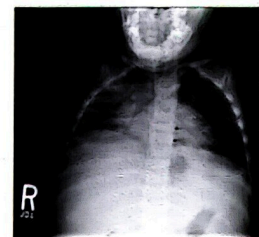
1. Extrahepatic biliary atresia (EHBA) : D/t **severe fibrosis.**
  - **m/c cause** of neonatal cholestasis.
2. Choledochal cyst : Cystic dilatation of bile duct.

**EVALUATION :**

Thorough history &amp; clinical examination.

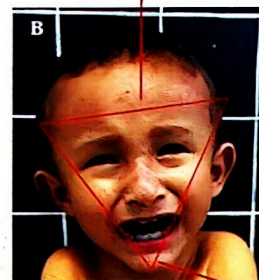
Serum bilirubin :

Total &amp; conjugated.

**Radiological imaging :** Primarily useful in EHBA.

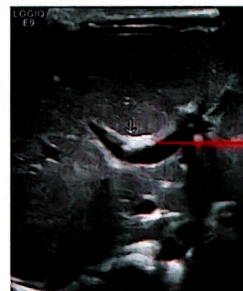
vertebral defects

Broad forehead



Triangular facies

Narrow jaw



Hyperechogenic

Triangular cord sign : Atritic bile duct

----- Active space -----

## Screening tests :

## 1. USG :

- microgallbladder/gallbladder ghost.
- Atretic bile duct (Hyperechogenic).

## 2. HIDA Scan : (Hepatobiliary imino diacetic scan) :

- ↓ excretion of the tracer (Imino diacetic).
- High negative predictive value.

## Biopsy :

**Best investigation :** Helps differentiate intrahepatic & extrahepatic cholestasis.

|                                        | Intrahepatic neonatal cholestasis | Extrahepatic biliary atresia |
|----------------------------------------|-----------------------------------|------------------------------|
| Hepatic architecture (Lobules)         | Damaged                           | Normal                       |
| Giant cells                            | +++                               | +                            |
| Portal reaction                        | Inflammation                      | Fibrosis                     |
| Neoductular/periductular proliferation | -                                 | +                            |

## MANAGEMENT :

## Medical management :

| Clinical impairment                                     | management                                                                                                                                                                    |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| malabsorption of dietary long chain triglycerides       | medium chain triglyceride oil : Helps absorb triglycerides                                                                                                                    |
| Fat soluble vitamins                                    | Vitamin A, D, E, K supplementation                                                                                                                                            |
| micronutrient deficiency                                | Cu, mg, Zn, $PO_4^{3-}$ supplementation                                                                                                                                       |
| Deficiency of water soluble vitamins                    | Supplement with twice the RDA with multivitamin syrup                                                                                                                         |
| Retention of biliary constituents<br>(Eg : Cholesterol) | <ul style="list-style-type: none"> <li>• ursodeoxycholic acid (UDCA) → ↓ fat absorption.</li> <li>• Cholestyramine (Bile acid binding resin → Promotes excretion).</li> </ul> |

## Surgical management :

## EHBA :

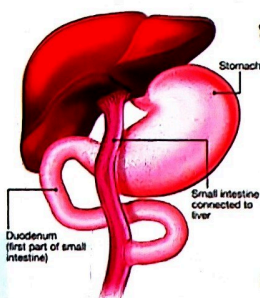
## 1. Kasai procedure :

- Connects liver to jejunum bypassing bile duct : **Hepatoportoenterostomy**.
- Good outcome if performed within 60 days of birth.

## 2. Definitive treatment : Transplant (D/t ↑ risk of cirrhosis &amp; fibrosis later in life).

## Choledochal cyst :

Excision of the cyst → Hepaticojejunostomy.



Kasai procedure

## Metabolic Liver Disease

00:45:17

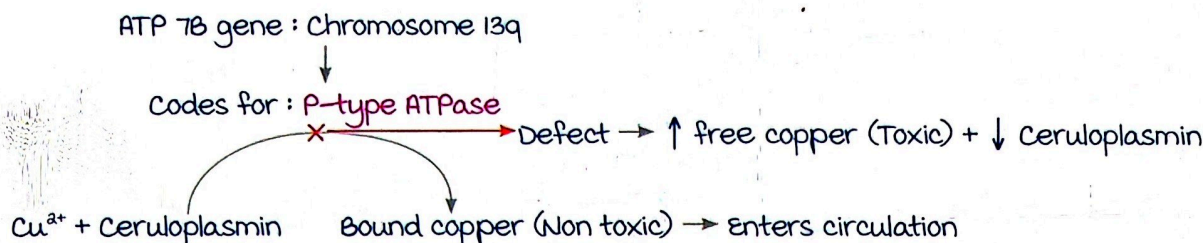
----- Active space -----

648c85cfee3b03a74e182fab

## WILSON DISEASE :

Autosomal recessive.

## Pathogenesis :



## Features :

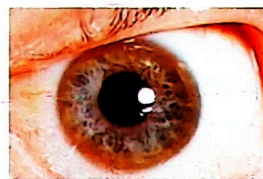
Onset : &gt; 3 yrs.

Liver : Commonly seen in children.

- Chronic liver disease → Cirrhosis → End stage liver disease.

CNS : Commonly seen in adults.

- Tremor, seizures.
  - Extrapyrmidal symptoms.
  - Psychosis & depression.
- } D/t basal ganglia involvement



KF ring

## Eye :

- $\text{Cu}^{2+}$  deposit in descemet's membrane of cornea → **Kayser fleischer (KF) ring**.
- **Sunflower cataract**.

Copper mediated hemolysis : **Coombs -ve**.

Arthralgia.

## Investigations :

## Screening :

- urine  $\text{Cu}^{2+}$  ↑
- Serum ceruloplasmin ↓

Gold standard : **Liver biopsy**.

- ↑  $\text{Cu}^{2+}$  deposition > 250  $\mu\text{g/g}$  of dry weight of the liver.
- Diagnostic of wilson disease.
- macrovesicular steatosis.

## Treatment :

## 1. Copper restriction :

- Nuts & chocolates.
- Animal products such as liver.

## 2. Copper chelator :

- **D-penicillamine**.
- Trientine.

## 3. Zinc : For long term maintenance treatment.

↓  $\text{Cu}^{2+}$  absorption from intestine.

## 4. Liver transplant :

- Definitive management.

----- Active space -----

**LIVER GLYCOGEN STORAGE DISORDERS (GSD) :****Pathogenesis :**Fasting state : Glycogen in liver  $\rightarrow$  Broken down to glucoseDefect  $\rightarrow$  GSD**Clinical presentation :**

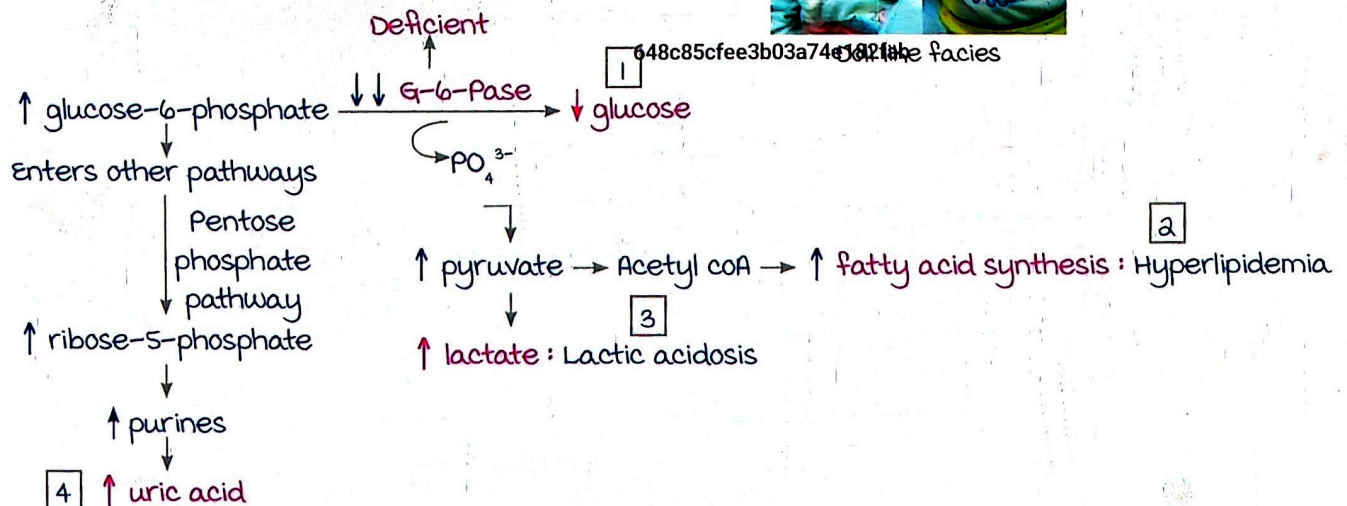
- Recurrent fasting hypoglycemia.
- Hepatomegaly (D/t glycogen accumulation).
- muscle involvement in certain subtypes : Exertional fatigue (D/t  $\downarrow$  ATP production).

**Types of liver GSD :**

| Type                | Deficiency of enzyme             | Features                                                                                                                                                                                 |
|---------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                   | Glycogen synthase                | No glycogen deposit in liver                                                                                                                                                             |
| I<br>(VON GIERKE)   | Glucose-6-phosphatase (G 6 Pase) | m/c type                                                                                                                                                                                 |
| II<br>(CORI'S)      | Debranching enzyme               | Can have muscle involvement                                                                                                                                                              |
| III<br>(ANDERSON'S) | Branching enzyme                 | <ul style="list-style-type: none"> <li>• 2<sup>nd</sup> m/c.</li> <li>• Can have muscle involvement.</li> <li>• Early onset cirrhosis <math>\rightarrow</math> Worst outcome.</li> </ul> |
| VI<br>(HER'S)       | Liver phosphorylase              | -                                                                                                                                                                                        |
| IX                  | Phosphorylase Kinase             | -                                                                                                                                                                                        |

**Other features in von Gierke's disease :**

1. Prominent cheeks  $\rightarrow$  Doll like facies.
2. Renomegaly on USG.
3. Biochemical abnormalities :



Differences b/w type I and type III GSD :

----- Active space -----

|                           | Type I (von Gierke)                                                                      | Type III (Cori)                                                                            |
|---------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| muscle involvement (Eye)  | -                                                                                        | +                                                                                          |
| Organomegaly              | <ul style="list-style-type: none"> <li>• Hepatomegaly.</li> <li>• Renomegaly.</li> </ul> | <ul style="list-style-type: none"> <li>• Hepatomegaly.</li> <li>• Splenomegaly.</li> </ul> |
| Serum lactate & uric acid | ↓↓↓                                                                                      | -                                                                                          |

### management :

1. Frequent feeds : Every 2-3 hrs, to prevent hypoglycemia.
2. Uncooked corn starch : Slow release of glucose over many hours  
 ↓  
 maintains stable blood glucose levels.
3. Night feeds :
  - Nasogastric feeds : Young children
  - **waxy maize heat modified starch** : Older children
 } Prevents hypoglycemia & ketosis overnight